# Fair Cooperation

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# Motivation

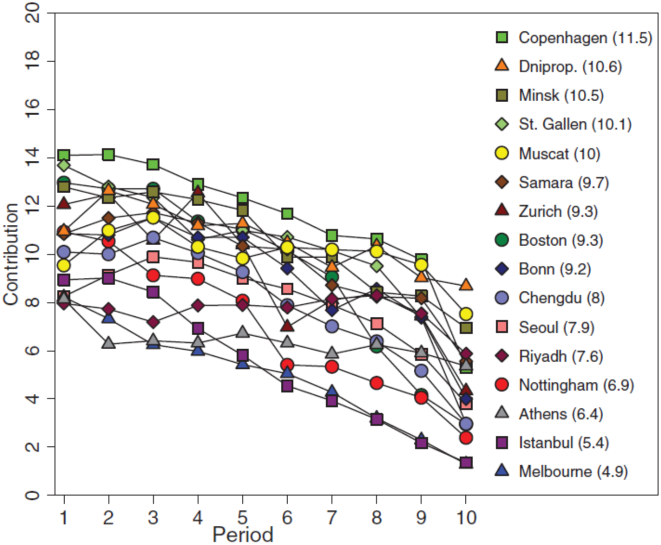
- Cooperation is a fundamental part of daily life
- We make a choice to cooperate in various different areas:
  - Some examples: teamwork, contributions to clubs or schools, environmental protection
- Cooperation is also a key building block in evolutionary biology

  *"Without cooperation there can be neither construction nor complexity in evolution"*

  *Nowak and Highfield (2012)*

# Cooperation Variation Across Societies

## Herrmann et al. (2008)



**Inequality Aversion**

- Preferences over unequal outcomes and fairness perceptions (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000; Almås et al., 2020, 2022; Cappelen et al., 2023).
- Perceptions of the source of inequality shape redistribution preferences (Almås et al., 2020; Cappelen et al., 2023).

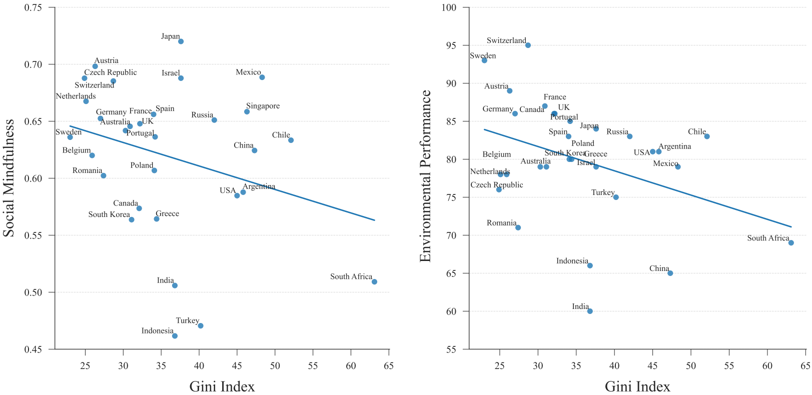
**Reciprocity**

- Cooperation depends on others' behavior and contributions (Fischbacher et al., 2001; Gächter, 2006).
- Conditional cooperation varies across societies (Kocher et al., 2008; Guiso et al., 2016; Rustagi, 2024).

Related factors include endowments, productivity, source of inequality, redistribution, and social networks (Hauser et al., 2019; Bland et al., 2023; Malthouse et al., 2023; Stallen et al., 2023; Cartwright et al., 2025).

# Pro-Sociality & Inequality: Some Evidence

## Van Doesum et al. (2021)




Gini Index from CIA, with associations to *Social Mindfulness* (p = 0.070) and *Environmental Performance* (p = 0.045).

# Overview

## Research Questions

1. Do economic and institutional structures shape cooperation?
2. Do perceptions of fairness and inequality affect cooperation?
   - If so, what behavioral channels may explain these effects?

## Contribution

- Use experimental variation in these structures to identify their causal effects on cooperation
- Show how these effects are consistent with inequality aversion and reciprocity

# Experimental Design: Overview

Between-subjects design with two stages, in groups of 4:

## Stage 1

- Individually complete a task based on ability and effort
- Earnings are redistributed within the group
- Treatment variation on difficulty and redistribution regime

## Stage 2

- Play a public good game
- First one-shot (no feedback), then eight rounds with feedback

## Implementation Details

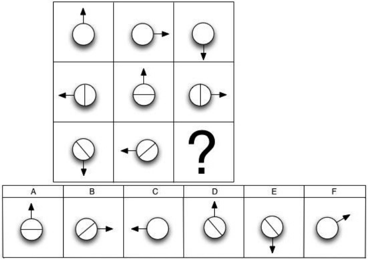
- Demographics and personality traits collected at the end
- 640 Prolific participants, equally distributed across treatments
- Pre-registered study: AEARCTR-0013588

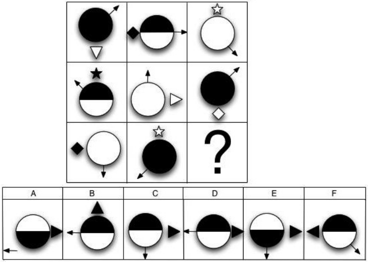
- Five matrix reasoning and five three-dimensional rotation questions (Condon and Revelle, 2014)
  - Receive 10 experimental tokens per correct answer
- After all group members complete, earnings are redistributed according to treatments

### Treatments: 2×2 Between-Subjects Design

| | | *Redistribution Rate* | |
|---|---|---|---|
| | | *Low* (5%) | *High* (50%) |
| *Task Difficulty* | *Difficult* | DLR | DHR |
| | *Easy* | ELR | EHR |

# Examples of Task Questions




▸ Population data ▸ Psychometric analysis

# Feedback Screen for Stage 1

**Part 1 Results Summary**

You are Player **B.**

| Group Earnings | | | | |
|---|---|---|---|---|
| A | B | C | D | Total |
| 10 | 40 | 0 | 20 | 70 |

| Your Payoff Details | | | |
|---|---|---|---|
| Earning | Tax Amount Paid | Group Redistribution | Your Payoff |
| 40 | 20 | 8.75 | 28.75 |

Note:

Your Payoff $=$ Your Earning $-$ Your Tax Amount Paid $+$ Group Redistribution

Your Tax Amount Paid $= \mathbf{50\%} \times$ Your Earning

Group Redistribution $= \frac{\mathbf{50\%} \times \text{Total Group Earning}}{\mathbf{4}}$

- Indicate the perceived fairness of the outcome after Stage 2

Screen

# Experimental Design: Stage 2

- Same groups play a public good game
  1. A one-shot round without feedback
  2. 8 repeated rounds with feedback
- Each round is identical in
  - Endowment of 100 tokens
  - Marginal Per Capita Return (MPCR) = 0.4
  - Individual $i$'s payoff function:

$$\pi_i(c_i) = 100 - c_i + 0.4 * \sum_{j=1}^{4} c_j$$

- Payment based on the one-shot outcome and one randomly selected repeated round

▸ Stage 2: Instructions

**Part 3 Results Summary**

You are Player **B.**

| Group Investments | | | | | |
|---|---|---|---|---|---|
| Round | A | B | C | D | Total |
| 1 | 89 | 45 | 34 | 67 | 235 |
| 2 | 67 | 23 | 89 | 45 | 224 |
| 3 | 89 | 67 | 23 | 89 | 268 |
| 4 | 50 | 50 | 50 | 50 | 200 |
| 5 | 50 | 50 | 50 | 50 | 200 |

| Your Payoff Details | | |
|---|---|---|
| Investment | Project Return | Your Payoff |
| 45 | 94 | 149 |
| 23 | 89.6 | 166.6 |
| 67 | 107.2 | 140.2 |
| 50 | 80 | 130 |
| 50 | 80 | 130 |

Note:

$\text{Your Payoff} = 100 - \text{Your Investment} + \text{Your Project Return}$

$\text{Your Project Return} = \mathbf{0.4} \times \text{Total Group Investment}$

# Sample Descriptive Statistics

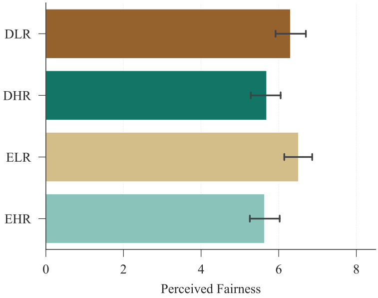
| | Difficult | | Easy | | Overall | |
|---|---|---|---|---|---|---|
| Variable | LR | HR | LR | HR | Mean | P-value |
| Pre-Redistribution Earnings | 47.375 | 46.438 | 62.125 | 55.062 | 52.750 | $< 0.001$ |
| | (1.995) | (2.005) | (2.036) | (2.193) | (1.058) | |
| Age | 38.594 | 38.025 | 37.469 | 39.062 | 38.288 | 0.743 |
| | (0.957) | (1.005) | (0.934) | (1.004) | (0.487) | |
| Female | 0.425 | 0.444 | 0.481 | 0.438 | 0.447 | 0.768 |
| | (0.039) | (0.039) | (0.040) | (0.039) | (0.020) | |
| Education | 2.706 | 2.562 | 2.706 | 2.725 | 2.675 | 0.628 |
| | (0.096) | (0.092) | (0.090) | (0.092) | (0.046) | |
| Groups | 40 | 40 | 40 | 40 | 160 | |
| Observations | 160 | 160 | 160 | 160 | 640 | |

Note: *Education*: 0 = None, 1 = High School, 2 = Technical College, 3 = Undergraduate, 4 = Graduate, 5 = Doctorate. Standard errors are reported in parentheses. P-values are based on Kruskal-Wallis tests.

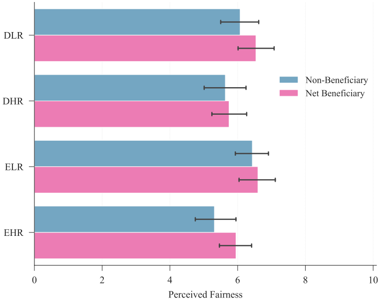
# Results

# Perceived Fairness

## Mostly judged according to redistribution - independent of beneficiary status



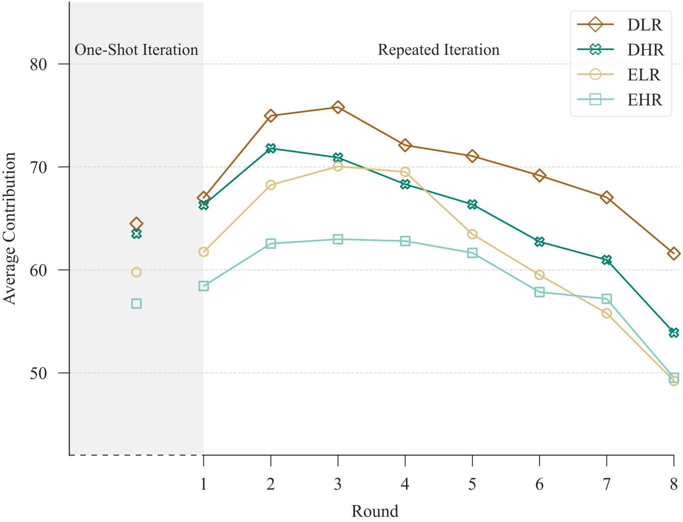

a) By regime




b) By beneficiary status

▸ Fairness Distributions

# Contributions by Treatment



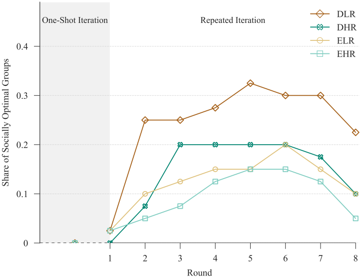

▸ Distributions

# Treatment Effects on Contribution

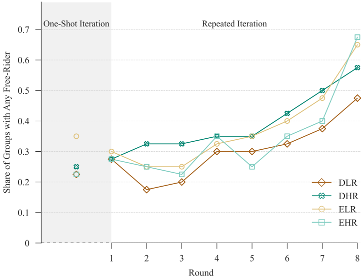
| | Contribution | | | | |
|---|---|---|---|---|---|
| | Overall | | One-Shot Game | Repeated Game | |
| | | | | First Half | All |
| | (1) | (2) | (3) | (4) | (5) |
| DLR | 10.389** | 9.892** | 7.746** | 10.787** | 10.161** |
| | (4.961) | (4.794) | (3.254) | (4.559) | (5.149) |
| DHR | 6.122 | 6.358 | 6.990** | 8.053* | 6.280 |
| | (4.594) | (4.428) | (3.312) | (4.321) | (4.718) |
| ELR | 3.068 | 2.376 | 2.420 | 5.571 | 2.370 |
| | (4.671) | (4.714) | (3.438) | (4.586) | (5.041) |
| Perceived Fairness | | 1.766*** | 1.021** | 1.014** | 1.859*** |
| | | (0.455) | (0.517) | (0.483) | (0.478) |
| Gini Index | | 0.092 | -0.124 | 0.099 | 0.119 |
| | | (0.154) | (0.115) | (0.149) | (0.164) |
| Pre-Redistribution Earnings | | 0.010 | 0.012 | 0.028 | 0.010 |
| | | (0.048) | (0.051) | (0.050) | (0.050) |
| Task Time | | -0.547 | -0.909** | -0.438 | -0.502 |
| | | (0.429) | (0.417) | (0.448) | (0.448) |
| Round | | -0.968*** | | 1.498*** | -1.664*** |
| | | (0.299) | | (0.491) | (0.327) |
| Constant | 58.860*** | 58.557*** | 62.844*** | 55.139*** | 61.728*** |
| | (3.323) | (8.900) | (8.480) | (9.131) | (9.421) |
| Controls | No | Yes | Yes | Yes | Yes |
| Observations | 5760 | 5760 | 640 | 2560 | 5120 |

Note: In all models, the baseline is the EHR treatment. Controls include age, gender, and education. Standard errors clustered at the group level are presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

# Group Outcomes by Treatment



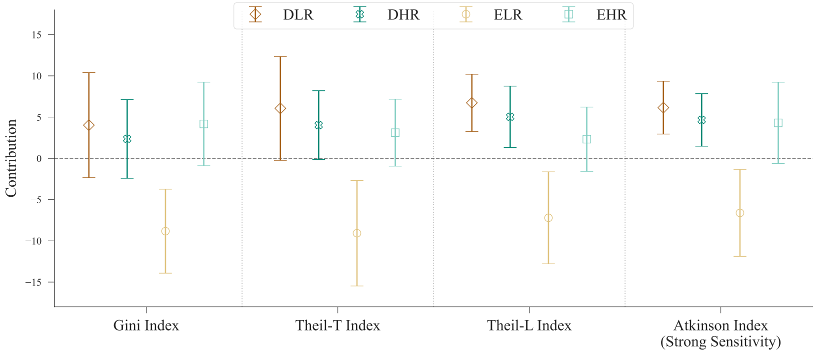

a) Socially optimal groups




b) Groups with any free-rider

Regression Analysis

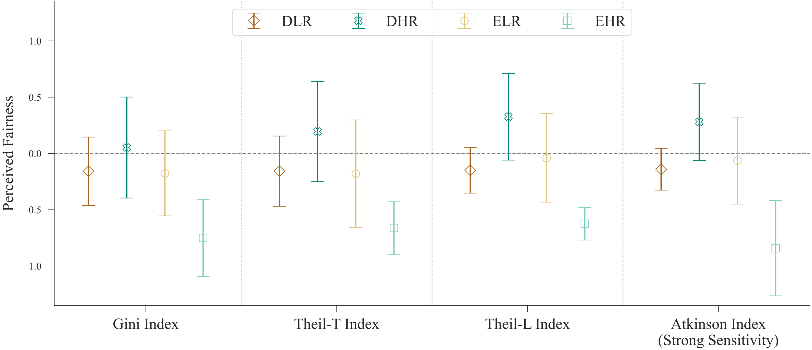
# Marginal Effects of Inequality on Contribution




Note: Average marginal effects of various standardised inequality indices on contributions, estimated using a random-effects GLS regression controlling for relevant covariates, with error bars indicating 95% confidence intervals.

▸ Measure Definitions ▸ Means ▸ Distributions ▸ Regression Analysis

# Marginal Effects of Inequality on Fairness Perception




Note: Average marginal effects of various standardised inequality indices on perceived fairness of the outcomes in the task, estimated using a random-effects GLS regression controlling for relevant covariates, with error bars indicating 95% confidence intervals.

Regression Analysis

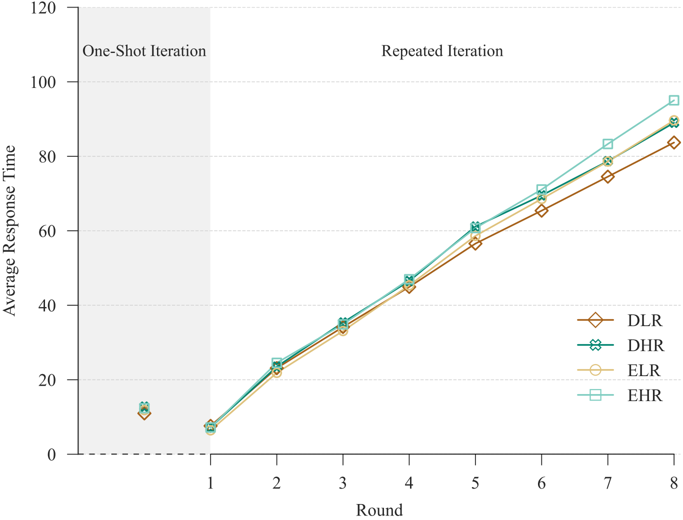
# Marginal Effects of Inequality on Contribution

## By Beneficiary Status


a) Non-Beneficiaries ($n = 305$)


b) Net Beneficiaries ($n = 335$)

# Response Times by Treatment




Distributions Regression Analysis

# Conclusion

1. Institutional regimes shape cooperation
   - More demanding environments generate stronger cooperation and higher socially optimal group outcomes.
   - Responses to redistribution differ across beneficiary groups, reflecting the roles of inequality aversion and reciprocity
2. Fairness perceptions matter for cooperation
   - Primarily shaped by the extent of redistribution
   - High inequality under high redistribution can be accepted in more demanding regimes

# Thanks for Your Cooperation and Attention - Questions?

## Some comments from our participants

*"Great game experience - enjoyed it a lot and believe it will inform much abut human nature, behaviour in decision instances, and attitude towards economic rewards"*

*"To see how you plan things when working in a team - do you look out for yourself, or the greater benefit of the group."*

# Appendix

# Implementation Details

Back

- Pre-registered: AEARCTR-0013588
- Dates: June 19–21, 2024
- Platform: Online experiment conducted on Qualtrics
- Participants: 640 participants recruited via Prolific
  - Randomized into four treatment conditions, with 40 groups of 4 participants each (160 participants per condition)
  - To ensure data quality we recruit participants with at least 99% approval rating and included two attention checks
  - UK sample
- Incentives: £1 + [£0, £10.8]
  - Conversion rate: 50 tokens= £1
  - Average payout: £7.59
- Average duration: 35.76 minutes

Summary Stats

# Sample Descriptive Statistics

Back

| | Difficult | | Easy | | Overall | |
|---|---|---|---|---|---|---|
| Variable | LR | HR | LR | HR | Mean | P-value |
| Age | 38.594 | 38.025 | 37.469 | 39.062 | 38.288 | 0.743 |
| | (0.957) | (1.005) | (0.934) | (1.004) | (0.487) | |
| Female | 0.425 | 0.444 | 0.481 | 0.438 | 0.447 | 0.768 |
| | (0.039) | (0.039) | (0.040) | (0.039) | (0.020) | |
| Education | 2.706 | 2.562 | 2.706 | 2.725 | 2.675 | 0.628 |
| | (0.096) | (0.092) | (0.090) | (0.092) | (0.046) | |
| Pre-Redistribution Earnings | 47.375 | 46.438 | 62.125 | 55.062 | 52.750 | $< 0.001$ |
| | (1.995) | (2.005) | (2.036) | (2.193) | (1.058) | |
| Task Time | 7.991 | 7.463 | 7.488 | 7.045 | 7.497 | 0.015 |
| | (0.227) | (0.246) | (0.257) | (0.245) | (0.122) | |
| Groups | 40 | 40 | 40 | 40 | 160 | |
| Observations | 160 | 160 | 160 | 160 | 640 | |

Note: Standard errors are reported in parentheses. P-values are based on Kruskal-Wallis tests.

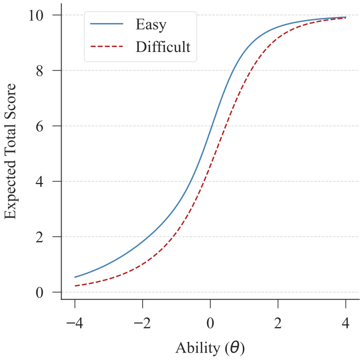
# Prior Psychometric Data of the Task

Back

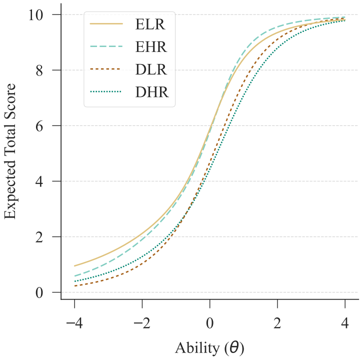
| Question | Easy | | | Difficult | | |
|---|---|---|---|---|---|---|
| | Accuracy | | Obs. | Accuracy | | Obs. |
| | Mean | SD | | Mean | SD | |
| 1 | 0.74 | 0.44 | 81,787 | 0.49 | 0.50 | 101,810 |
| 2 | 0.61 | 0.49 | 80,701 | 0.41 | 0.49 | 79,801 |
| 3 | 0.56 | 0.50 | 80,748 | 0.40 | 0.49 | 100,621 |
| 4 | 0.56 | 0.50 | 80,530 | 0.27 | 0.45 | 99,874 |
| 5 | 0.54 | 0.50 | 80,409 | 0.29 | 0.45 | 99,911 |
| 6 | 0.39 | 0.49 | 40,523 | 0.39 | 0.49 | 40,523 |
| 7 | 0.38 | 0.49 | 3,305 | 0.38 | 0.49 | 3,305 |
| 8 | 0.37 | 0.48 | 3,551 | 0.24 | 0.43 | 3,654 |
| 9 | 0.39 | 0.49 | 3,361 | 0.27 | 0.45 | 3,435 |
| 10 | 0.40 | 0.49 | 3,455 | 0.21 | 0.40 | 3,573 |
| Overall | 0.49 | | | 0.34 | | |

Note: Source from Dworak et al. (2021)

# Psychometric Data in the Experiment ▸ Back




By production difficulty




By treatment condition

Note: Test characteristic curves estimated from two-parameter (difficulty and discrimination) logistic item response theory models. Panel (a) plots expected total scores as a function of latent ability ($\theta$) for the Easy and Difficult conditions. Panel (b) presents the corresponding curves for each of the four treatment groups.

## Part 1 Instructions

In this part, your task is to answer **10** questions. Five of them are about matrix reasoning and the other five are about three-dimensional rotation.

There is a correct answer to each question. For **every correct** response you submit, you will earn **10** experimental tokens. This scheme applies to all group members. You have a maximum of 15 minutes to complete this task.

After all members complete the task, **a tax amount of 50%** of each member's earnings will be collected, pooled, and **evenly distributed** among all four members of the group. For example, if 100 tokens are earned in the task, the tax amount to be paid is 50 tokens. This amount is evenly divided among the group members, meaning that each member receives a redistribution payment of 12.5 tokens.

Your payout in this part is:

$$\text{Your Earning} - \mathbf{50\%} \times \text{Your Earning} + \frac{\mathbf{50\%} \times \text{Total Group Earning}}{\mathbf{4}}$$

# Feedback Screen for Stage 1

Back

**Part 1 Results Summary**

You are Player **B.**

| Group Earnings | | | | |
|---|---|---|---|---|
| A | B | C | D | Total |
| 10 | 40 | 0 | 20 | 70 |

| Your Payoff Details | | | |
|---|---|---|---|
| Earning | Tax Amount Paid | Group Redistribution | Your Payoff |
| 40 | 20 | 8.75 | 28.75 |

Note:

$$\text{Your Payoff} = \text{Your Earning} - \text{Your Tax Amount Paid} + \text{Group Redistribution}$$

$$\text{Your Tax Amount Paid} = \mathbf{50\%} \times \text{Your Earning}$$

$$\text{Group Redistribution} = \frac{\mathbf{50\%} \times \text{Total Group Earning}}{\mathbf{4}}$$

## Part 2 Instructions

In this part, you will engage in a game with your group.

Each member starts with **an endowment of 100** units of experimental tokens. Your task is to **invest any amount** in a "group project". You can invest any amount **between 0 and 100**, including 0 and 100. Every group member makes such a decision.

After all individual investment decisions are made, the group's total investment will be **multiplied by 1.6**. The resulting collective return will then be **evenly distributed** among all four members of the group. For example, if 100 tokens are invested in the project, the project's collective return increases to 160 tokens. This amount is evenly divided among the group members, so each member receives a project return of 40 tokens.

Your payout in this part is:

$$100 - \text{Your Investment} + \mathbf{0.4} \times \text{Total Group Investment}$$

The outcome for this part and your corresponding payout will be listed together with the payout from the other parts at the end of the experiment.

# Stage 2- Contribution Screen

Back

You have received 100 units.

Please indicate in the box below how many units to invest in the group project.

Your **investment** in the group project (0 - 100):

# Stage 1 - Instructions ▸ Back

### Part 3 Instructions

In this part, you will engage in the same game as in Part 2. This game will be repeated across eight rounds.

Your payoff for each round will be displayed after the round concludes. At the end of the experiment, one of the eight rounds from this part will be randomly selected, with each round having an equal chance of being chosen. Your payout for this part will be determined by your group's decisions in the round that is randomly selected.

To remind you the rules of the game:

1) In each round, each member has an endowment of **100** units of experimental tokens.

2) Your task is to invest any amount in a "group project". You can invest any amount **between 0 and 100**, including 0 and 100. Every group member makes such a decision.

3) After all individual investment decisions are made, the group's total investment will be multiplied by **1.6**. The resulting amount will then be **evenly distributed** among all four members of the group. For example, if 100 tokens are invested in the project, the project's collective return increases to 160 tokens. This amount is evenly divided among the group members, so each member receives a project return of 40 tokens.

4) Your payoff in the game for each round is:

$$100 - \text{Your Investment} + \mathbf{0.4} \times \text{Total Group Investment}$$

# Feedback Screen for Stage 2

▸ Back

**Part 3 Results Summary**

You are Player **B.**

| Group Investments | | | | | |
|---|---|---|---|---|---|
| Round | A | B | C | D | Total |
| 1 | 89 | 45 | 34 | 67 | 235 |
| 2 | 67 | 23 | 89 | 45 | 224 |
| 3 | 89 | 67 | 23 | 89 | 268 |
| 4 | 50 | 50 | 50 | 50 | 200 |
| 5 | 50 | 50 | 50 | 50 | 200 |

| Your Payoff Details | | |
|---|---|---|
| Investment | Project Return | Your Payoff |
| 45 | 94 | 149 |
| 23 | 89.6 | 166.6 |
| 67 | 107.2 | 140.2 |
| 50 | 80 | 130 |
| 50 | 80 | 130 |

Note:

$\text{Your Payoff} = 100 - \text{Your Investment} + \text{Your Project Return}$

$\text{Your Project Return} = \mathbf{0.4} \times \text{Total Group Investment}$

# Fairness Perception ▸ Back

Now, we would like to understand your perception of fairness regarding your outcomes in Part 1.

A rating of **0** indicates that you perceive the outcome as **completely unfair**, while a rating of **10** indicates that you view the outcome as **perfectly fair.** Please select a number between 0 and 10 that best reflects your feelings about your outcomes in Part 1. Below is a summary of the results from Part 1.

**Part 1 Results Summary**

You are **Player B**.

| Group Earnings | | | | |
|---|---|---|---|---|
| A | B | C | D | Total |
| 10 | 40 | 0 | 20 | 70 |

| Your Payoff Details | | | |
|---|---|---|---|
| Earning | Tax Amount Paid | Group Redistribution | Your Payoff |
| 40 | 20 | 8.75 | 28.75 |

Note:

$\text{Your Payoff} = \text{Your Earning} - \text{Your Tax Amount Paid} + \text{Group Redistribution}$

$\text{Your Tax Amount Paid} = \mathbf{50\%} \times \text{Your Earning}$

$\text{Group Redistribution} = \frac{\mathbf{50\%} \times \text{Total Group Earning}}{\mathbf{4}}$

How fair do you perceive your outcomes in Part 1 to be?

**Very unfair** **Very fair**

0 1 2 3 4 5 6 7 8 9 10

**Fairness**

# Sample Descriptive Statistics

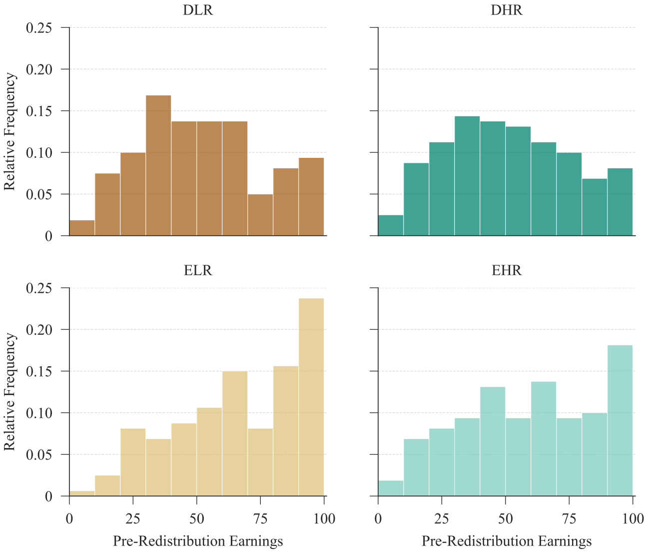
Back

| | Difficult | | Easy | | Overall | |
|---|---|---|---|---|---|---|
| Variable | LR | HR | LR | HR | Mean | P-value |
| Age | 38.594 | 38.025 | 37.469 | 39.062 | 38.288 | 0.743 |
| | (0.957) | (1.005) | (0.934) | (1.004) | (0.487) | |
| Female | 0.425 | 0.444 | 0.481 | 0.438 | 0.447 | 0.768 |
| | (0.039) | (0.039) | (0.040) | (0.039) | (0.020) | |
| Education | 2.706 | 2.562 | 2.706 | 2.725 | 2.675 | 0.628 |
| | (0.096) | (0.092) | (0.090) | (0.092) | (0.046) | |
| Pre-Redistribution Earnings | 47.375 | 46.438 | 62.125 | 55.062 | 52.750 | $< 0.001$ |
| | (1.995) | (2.005) | (2.036) | (2.193) | (1.058) | |
| Task Time | 7.991 | 7.463 | 7.488 | 7.045 | 7.497 | 0.015 |
| | (0.227) | (0.246) | (0.257) | (0.245) | (0.122) | |
| Groups | 40 | 40 | 40 | 40 | 160 | |
| Observations | 160 | 160 | 160 | 160 | 640 | |

Note: Standard errors are reported in parentheses. P-values are based on Kruskal-Wallis tests.

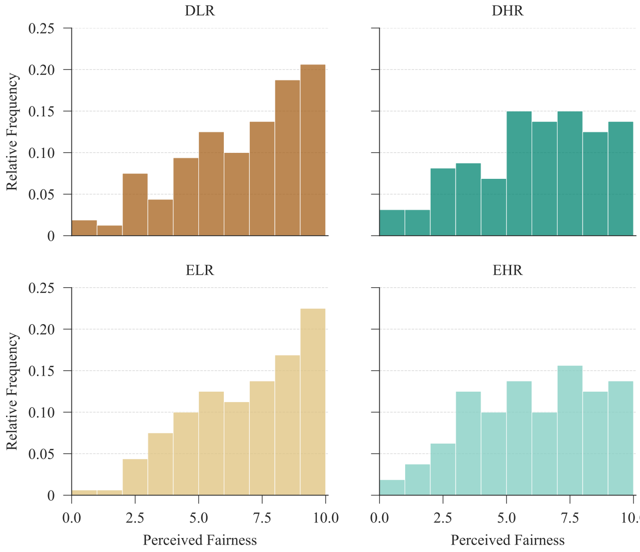
# Distributions of Pre-Redistribution Earnings by Treatment

Back



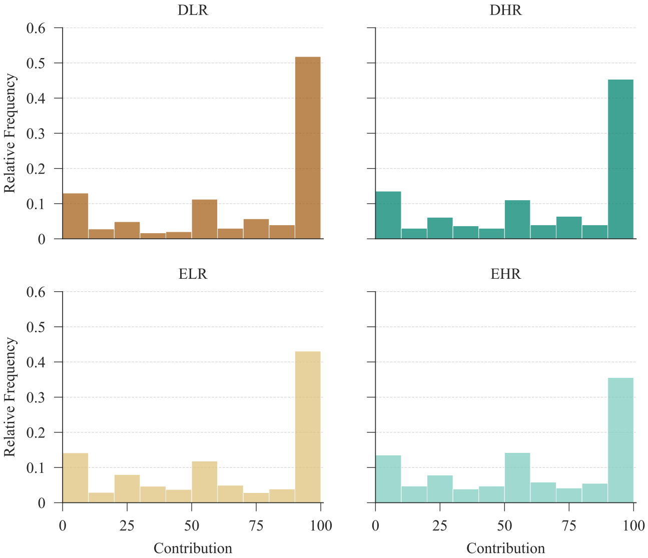
# Distributions of Perceived Fairness by Treatment

Back



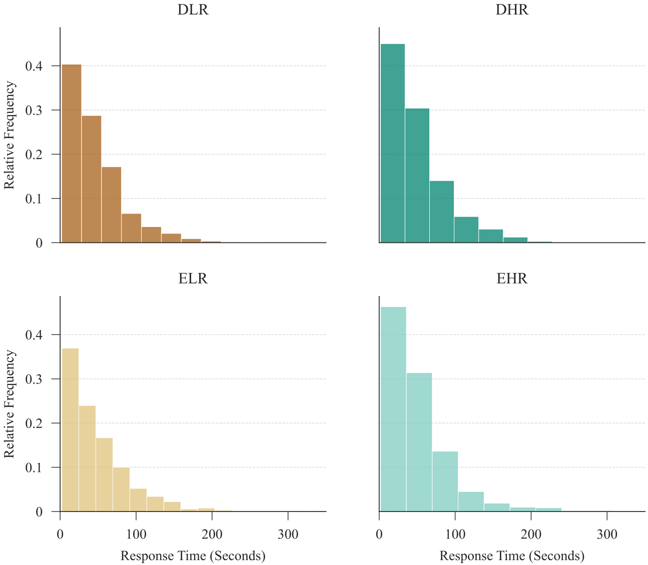
# Distributions of PPG Contribution by Treatment

Back



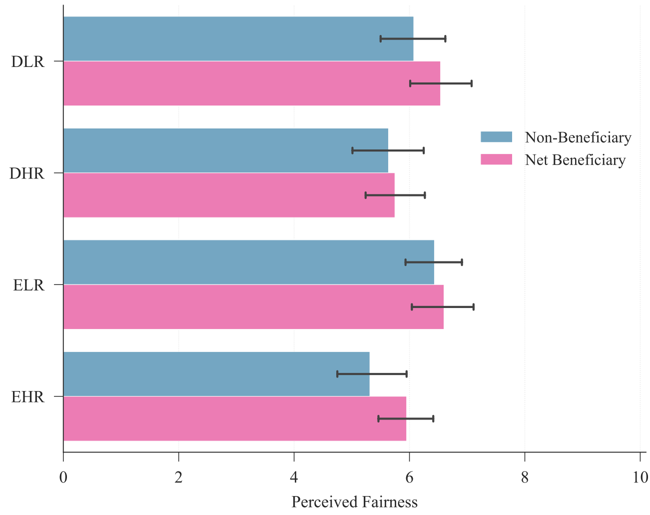
# Distributions of Response Time by Treatment

▸ Back



# Perceived Fairness by Benefiacry Status

Back



# Treatment Effects on Contribution

Back

| | Contribution | | | | |
|---|---|---|---|---|---|
| | Overall | | One-Shot Game | Repeated Game | |
| | | | | First Half | All |
| | (1) | (2) | (3) | (4) | (5) |
| DLR | 10.389** | 9.892** | 7.746** | 10.787** | 10.161** |
| | (4.961) | (4.794) | (3.254) | (4.559) | (5.149) |
| DHR | 6.122 | 6.358 | 6.990** | 8.053* | 6.280 |
| | (4.594) | (4.428) | (3.312) | (4.321) | (4.718) |
| ELR | 3.068 | 2.376 | 2.420 | 5.571 | 2.370 |
| | (4.671) | (4.714) | (3.438) | (4.586) | (5.041) |
| Perceived Fairness | | 1.766*** | 1.021** | 1.014** | 1.859*** |
| | | (0.455) | (0.517) | (0.483) | (0.478) |
| Gini Index | | 0.092 | -0.124 | 0.099 | 0.119 |
| | | (0.154) | (0.115) | (0.149) | (0.164) |
| Pre-Redistribution Earnings | | 0.010 | 0.012 | 0.028 | 0.010 |
| | | (0.048) | (0.051) | (0.050) | (0.050) |
| Task Time | | -0.547 | -0.909** | -0.438 | -0.502 |
| | | (0.429) | (0.417) | (0.448) | (0.448) |
| Round | | -0.968*** | | 1.498*** | -1.664*** |
| | | (0.299) | | (0.491) | (0.327) |
| Constant | 58.860*** | 58.557*** | 62.844*** | 55.139*** | 61.728*** |
| | (3.323) | (8.900) | (8.480) | (9.131) | (9.421) |
| Controls | No | Yes | Yes | Yes | Yes |
| Observations | 5760 | 5760 | 640 | 2560 | 5120 |

Note: In all models, the baseline is the EHR group. Controls include age, gender, and education. Standard errors clustered at the group level are presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

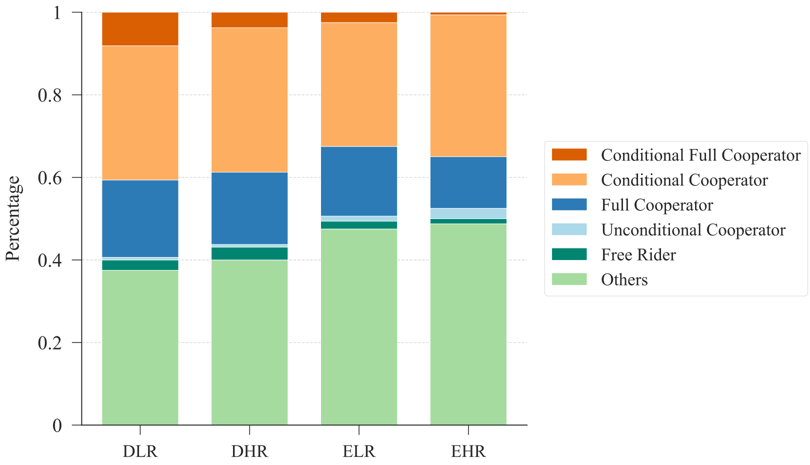
# Treatment Effects on Group Outcomes ▸ Back

| | Socially Optimal | | With Any Free-Rider | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| DLR | 3.043** | 3.322** | 0.835 | 1.010 |
| | (1.498) | (1.888) | (0.303) | (0.399) |
| DHR | 1.611 | 2.036 | 1.200 | 1.306 |
| | (0.843) | (1.227) | (0.404) | (0.456) |
| ELR | 1.375 | 0.904 | 1.186 | 1.184 |
| | (0.745) | (0.480) | (0.399) | (0.446) |
| Perceived Fairness | | 1.529*** | | 0.803** |
| | | (0.230) | | (0.080) |
| Gini Index | | 0.992 | | 0.976 |
| | | (0.024) | | (0.016) |
| Pre-redistribution Earnings | | 1.017 | | 1.011 |
| | | (0.017) | | (0.011) |
| Task Time | | 0.902 | | 0.992 |
| | | (0.075) | | (0.072) |
| Round | | 1.213*** | | 1.181*** |
| | | (0.030) | | (0.030) |
| Observations | 1440 | 1440 | 1440 | 1440 |

Note: Odd ratios from logistic regressions. The baseline for all models is the EHR group. Standard errors clustered at the group level are presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

# Cooperation Profiles by Treatment

Back



# Measuring Inequality ▸ Back

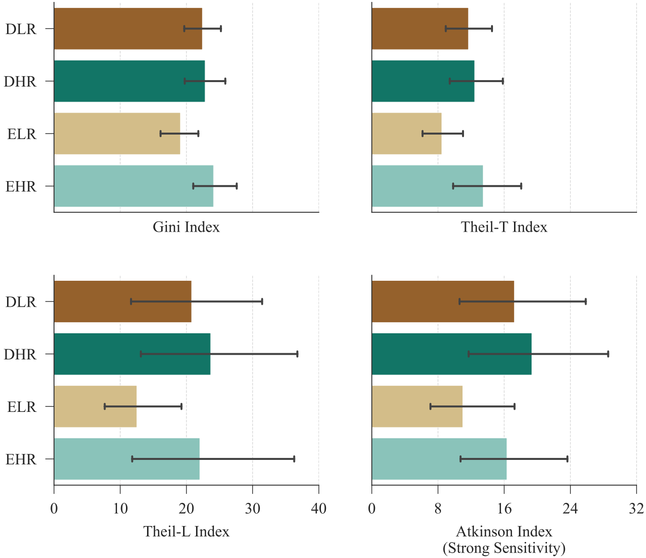
| Measure | Distributional Sensitivity | Formula |
|---|---|---|
| Gini Index | Uniform across distribution | $\frac{\sum_{i,j}\lvert x_i - x_j\rvert}{2n^2\mu}$ |
| Theil-T Index (entropy index) | Upper-end (top incomes) | $\frac{1}{n}\sum_{i=1}^{n}\frac{x_i}{\mu}\ln\left(\frac{x_i}{\mu}\right)$ |
| Theil-L Index (mean log deviation) | Lower-end (bottom incomes) | $\frac{1}{n}\sum_{i=1}^{n}\ln\left(\frac{\mu}{x_i}\right)$ |
| Atkinson Index $(0 \leq \epsilon \neq 1)$ | Lower-end, governed by $\epsilon$ | $1-\left(\frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i}{\mu}\right)^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}}$ |
| Atkinson Index $(\epsilon = 1)$ | Strong lower-end sensitivity | $1-\left(\prod_{i=1}^{n}\frac{x_i}{\mu}\right)^{\frac{1}{n}}$ |

Note: The Atkinson Index incorporates a parameter $\epsilon$ that represents the aversion to the inequality of the lower end.

# Inequality Across Treatments

Back

## Mean values of pre-redistribution inequality measures




Note: Error bars indicate the 95% CI.

# Marginal Effects of Inequality on Contribution

Back

## Non-Beneficiaries

# Marginal Effects of Inequality on Contribution

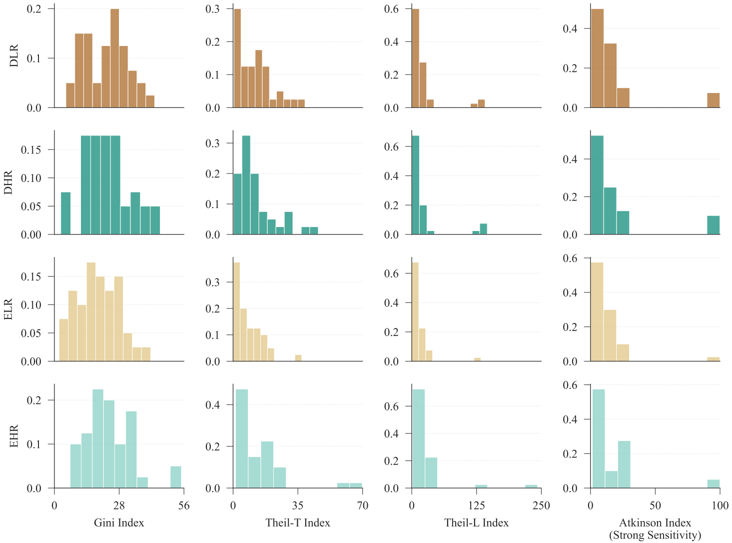
Back

## Net Beneficiaries

# Inequality Across Treatments

Back

## Distributions of pre-redistribution inequality measures



# Impact of Inequality on Fairness Perception ▸ Back

| | Perceived Fairness | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | Atkinson Index (Varying Sensitivities) | | | |
| | Gini Index | Theil-T Index | Theil-L Index | Mild | Moderate | Strong | Severe |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Inequality | -0.076*** | -0.062*** | -0.018*** | -0.213*** | -0.089*** | -0.037*** | -0.034*** |
| | (0.018) | (0.011) | (0.002) | (0.035) | (0.014) | (0.009) | (0.008) |
| LR $\times$ Inequality | 0.058** | 0.045* | 0.017*** | 0.159* | 0.070** | 0.034*** | 0.027** |
| | (0.025) | (0.025) | (0.006) | (0.084) | (0.034) | (0.012) | (0.012) |
| Difficult $\times$ Inequality | 0.081*** | 0.081*** | 0.027*** | 0.287*** | 0.124*** | 0.049*** | 0.043*** |
| | (0.028) | (0.023) | (0.006) | (0.078) | (0.031) | (0.012) | (0.012) |
| Difficult $\times$ LR $\times$ Inequality | -0.080** | -0.078** | -0.030*** | -0.285** | -0.127*** | -0.052*** | -0.041*** |
| | (0.038) | (0.036) | (0.009) | (0.122) | (0.048) | (0.015) | (0.016) |
| LR | -0.540 | 0.256 | 0.564* | 0.307 | 0.357 | 0.378 | 0.161 |
| | (0.584) | (0.366) | (0.310) | (0.354) | (0.340) | (0.323) | (0.368) |
| Difficult | -1.965*** | -1.072*** | -0.610* | -1.018*** | -0.943*** | -0.833** | -1.187*** |
| | (0.659) | (0.383) | (0.322) | (0.368) | (0.351) | (0.337) | (0.391) |
| Difficult $\times$ LR | 1.644* | 0.766 | 0.366 | 0.726 | 0.668 | 0.584 | 0.846 |
| | (0.900) | (0.570) | (0.451) | (0.546) | (0.516) | (0.469) | (0.551) |
| Constant | 7.702*** | 6.687*** | 6.176*** | 6.602*** | 6.504*** | 6.385*** | 6.715*** |
| | (0.651) | (0.518) | (0.496) | (0.511) | (0.505) | (0.509) | (0.539) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 640 | 640 | 640 | 640 | 640 | 640 | 640 |

Note: In all models, the baseline is the EHR group. Controls include age, gender, education, perceived fairness, task performance, and effort time. Standard errors clustered at the group level are presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

# Impact of Inequality on Contribution

Back

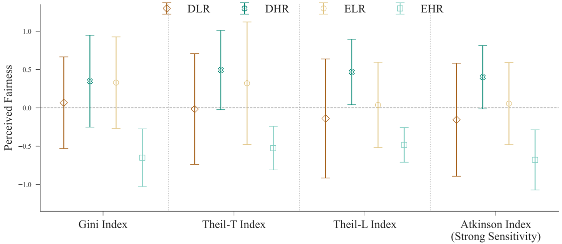
| | Contribution | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | Atkinson Index (Varying Sensitivities) | | | |
| | Gini Index | Theil-T Index | Theil-L Index | Mild | Moderate | Strong | Severe |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Inequality | 0.423 | 0.293 | 0.066 | 0.951 | 0.383 | 0.187* | 0.164 |
| | (0.262) | (0.195) | (0.057) | (0.673) | (0.284) | (0.110) | (0.116) |
| LR × Inequality | -1.318*** | -1.146*** | -0.273*** | -3.804*** | -1.504*** | -0.475*** | -0.419*** |
| | (0.371) | (0.365) | (0.101) | (1.266) | (0.525) | (0.163) | (0.160) |
| Difficult × Inequality | -0.183 | 0.086 | 0.078 | 0.475 | 0.247 | 0.016 | 0.027 |
| | (0.362) | (0.279) | (0.078) | (0.952) | (0.388) | (0.130) | (0.139) |
| Difficult × LR × Inequality | 1.486*** | 1.337*** | 0.322** | 4.477*** | 1.774*** | 0.540*** | 0.471** |
| | (0.559) | (0.518) | (0.126) | (1.732) | (0.682) | (0.193) | (0.202) |
| LR | 29.155*** | 13.025* | 5.971 | 11.655* | 10.121 | 8.090 | 11.393 |
| | (10.546) | (6.738) | (5.283) | (6.480) | (6.175) | (5.711) | (6.942) |
| Difficult | 10.804 | 5.387 | 4.180 | 4.752 | 4.381 | 5.300 | 5.234 |
| | (10.230) | (6.231) | (5.046) | (6.014) | (5.765) | (5.464) | (6.591) |
| Difficult × LR | -29.421* | -11.555 | -3.131 | -10.003 | -8.149 | -5.368 | -8.992 |
| | (15.622) | (10.097) | (7.672) | (9.621) | (9.040) | (8.121) | (9.731) |
| Constant | 49.775*** | 55.591*** | 58.073*** | 56.025*** | 56.418*** | 55.895*** | 54.753*** |
| | (11.014) | (8.454) | (7.776) | (8.316) | (8.185) | (8.149) | (8.863) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 5760 | 5760 | 5760 | 5760 | 5760 | 5760 | 5760 |

Note: In all models, the baseline is the EHR group and inequality indices are standardised. Controls include age, gender, education, perceived fairness, task performance, effort time, and round. Standard errors clustered at the group level are presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

# Marginal Effects of Inequality on Fairness Perception

## By Beneficiary Status


a) Non-Beneficiaries ($n = 305$)




b) Net Beneficiaries ($n = 335$)

# Treatment Effects on Response Time

Figure Conclusion

| | Response Time (1) | Feedback Time (2) | Contribution Time (3) |
|---|---|---|---|
| DLR | -7.139** | -7.621** | -1.211*** |
| | (2.893) | (3.465) | (0.450) |
| DHR | -3.569 | -3.546 | -0.811* |
| | (2.895) | (3.476) | (0.481) |
| ELR | -1.131 | -0.519 | -0.727* |
| | (2.742) | (3.324) | (0.381) |
| Fairness Score | 0.123 | 0.122 | 0.028 |
| | (0.340) | (0.413) | (0.054) |
| Gini index | 0.147 | 0.168 | 0.017 |
| | (0.092) | (0.111) | (0.016) |
| Pre-Redistribution Earnings | -0.203*** | -0.224*** | -0.029*** |
| | (0.034) | (0.040) | (0.006) |
| Task Time | 2.044*** | 2.341*** | 0.223*** |
| | (0.313) | (0.371) | (0.051) |
| Round | 10.667*** | 11.187*** | -0.587*** |
| | (0.304) | (0.340) | (0.046) |
| Constant | -19.925*** | -31.859*** | 8.920*** |
| | (6.110) | (7.637) | (0.887) |
| Controls | Yes | Yes | Yes |
| Observations | 5760 | 4480 | 5760 |

Note: In all models, the baseline is the EHR group. Controls include age, gender, and education. Standard errors clustered at the group level are presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.